In [23]:
```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
print(data.head())
```

```
           job_title  experience_years education_level  skills_count  \
0         AI Engineer                10        Bachelor             2
1        Data Analyst                 5        Bachelor            17
2  Frontend Developer                18             PhD             4
3    Business Analyst                19             PhD            13
4     Product Manager                15        Bachelor             7

        industry company_size   location remote_work  certifications  salary
0     Healthcare       Medium      India      Hybrid               2  109413
1        Telecom        Small  Australia          No               0   93764
2          Media       Medium  Singapore          No               1  148123
3         Retail       Medium     Canada         Yes               0  189123
4  Manufacturing        Large     Sweden         Yes               0  165069
```

In [24]:
```
print(data.columns)
```

```
Index(['job_title', 'experience_years', 'education_level', 'skills_count',
       'industry', 'company_size', 'location', 'remote_work', 'certifications',
       'salary'],
      dtype='str')
```

In [25]:
```
print(data.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   job_title         250000 non-null  str
 1   experience_years  250000 non-null  int64
 2   education_level   250000 non-null  str
 3   skills_count      250000 non-null  int64
 4   industry          250000 non-null  str
 5   company_size      250000 non-null  str
 6   location          250000 non-null  str
 7   remote_work       250000 non-null  str
 8   certifications    250000 non-null  int64
 9   salary            250000 non-null  int64
dtypes: int64(4), str(6)
memory usage: 19.1 MB
None
```

In [26]:
```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
print(data.describe(include='all'))
```

```
                job_title  experience_years education_level   skills_count  \
count              250000     250000.000000          250000  250000.000000
unique                 12               NaN               5            NaN
top     Backend Developer               NaN          Master            NaN
freq                21125               NaN           50352            NaN
mean                  NaN         10.005408             NaN       9.997812
std                   NaN          6.060602             NaN       5.479288
min                   NaN          0.000000             NaN       1.000000
25%                   NaN          5.000000             NaN       5.000000
50%                   NaN         10.000000             NaN      10.000000
75%                   NaN         15.000000             NaN      15.000000
max                   NaN         20.000000             NaN      19.000000

       industry company_size   location remote_work  certifications  \
count    250000       250000     250000      250000   250000.000000
unique       10            5         10           3             NaN
top     Finance        Large  Australia          No             NaN
freq      25393        50254      25258       83621             NaN
mean        NaN          NaN        NaN         NaN        2.491928
std         NaN          NaN        NaN         NaN        1.706475
min         NaN          NaN        NaN         NaN        0.000000
25%         NaN          NaN        NaN         NaN        1.000000
50%         NaN          NaN        NaN         NaN        2.000000
75%         NaN          NaN        NaN         NaN        4.000000
max         NaN          NaN        NaN         NaN        5.000000

               salary
count   250000.000000
unique            NaN
top               NaN
freq              NaN
mean    145718.080524
std      37407.952729
min      31867.000000
25%     119358.000000
50%     143453.000000
75%     169492.000000
max     333046.000000
```

In [27]:
```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
data.duplicated().sum()
```

Out[27]:
```
np.int64(0)
```

In [28]:
```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
data.isnull().sum()
```

Out[28]:
```
job_title           0
experience_years    0
education_level     0
skills_count        0
industry            0
company_size        0
location            0
remote_work         0
certifications      0
salary              0
dtype: int64
```

In [29]:
```
data.isnull().values.any()
```

Out[29]:
```
np.False_
```

In [30]:
```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
data['education_level'] = data['education_level'].astype('category').cat.codes
print(data.head())
```

```
           job_title  experience_years  education_level  skills_count  \
0         AI Engineer                10                0             2
1        Data Analyst                 5                0            17
2  Frontend Developer                18                4             4
3    Business Analyst                19                4            13
4     Product Manager                15                0             7

        industry company_size   location remote_work  certifications  salary
0     Healthcare       Medium      India      Hybrid               2  109413
1        Telecom        Small  Australia          No               0   93764
2          Media       Medium  Singapore          No               1  148123
3         Retail       Medium     Canada         Yes               0  189123
4  Manufacturing        Large     Sweden         Yes               0  165069
```

In [31]:
```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
data = pd.get_dummies(data,columns=['job_title','industry','company_size','locat
data = pd.get_dummies(data,columns=['remote_work'])
print(data.head())
```

```
   experience_years  skills_count  certifications  salary  \
0                10             2               2  109413
1                 5            17               0   93764
2                18             4               1  148123
3                19            13               0  189123
4                15             7               0  165069

   job_title_AI Engineer  job_title_Backend Developer  \
0                      1                            0
1                      0                            0
2                      0                            0
3                      0                            0
4                      0                            0

   job_title_Business Analyst  job_title_Cloud Engineer  \
0                           0                         0
1                           0                         0
2                           0                         0
3                           1                         0
4                           0                         0

   job_title_Cybersecurity Analyst  job_title_Data Analyst  \
0                                0                       0
1                                0                       1
2                                0                       0
3                                0                       0
4                                0                       0

   job_title_Data Scientist  job_title_DevOps Engineer  \
0                         0                          0
1                         0                          0
2                         0                          0
3                         0                          0
4                         0                          0

   job_title_Frontend Developer  job_title_Machine Learning Engineer  \
0                             0                                    0
1                             0                                    0
2                             1                                    0
3                             0                                    0
4                             0                                    0

   job_title_Product Manager  job_title_Software Engineer  \
0                          0                            0
1                          0                            0
2                          0                            0
3                          0                            0
4                          1                            0

   industry_Consulting  industry_Education  industry_Finance  \
0                    0                   0                 0
1                    0                   0                 0
2                    0                   0                 0
3                    0                   0                 0
4                    0                   0                 0

   industry_Government  industry_Healthcare  industry_Manufacturing  \
0                    0                   1                       0
1                    0                   0                       0
2                    0                   0                       0
```

```
3                    0                    0                       0
4                    0                    0                       1

   industry_Media  industry_Retail  industry_Technology  industry_Telecom  \
0               0                0                    0                 0
1               0                0                    0                 1
2               1                0                    0                 0
3               0                1                    0                 0
4               0                0                    0                 0

   company_size_Enterprise  company_size_Large  company_size_Medium  \
0                        0                   0                    1
1                        0                   0                    0
2                        0                   0                    1
3                        0                   0                    1
4                        0                   1                    0

   company_size_Small  company_size_Startup  location_Australia  \
0                   0                     0                   0
1                   1                     0                   1
2                   0                     0                   0
3                   0                     0                   0
4                   0                     0                   0

   location_Canada  location_Germany  location_India  location_Netherlands  \
0                0                 0               1                     0
1                0                 0               0                     0
2                0                 0               0                     0
3                1                 0               0                     0
4                0                 0               0                     0

   location_Remote  location_Singapore  location_Sweden  location_UK  \
0                0                   0                0            0
1                0                   0                0            0
2                0                   1                0            0
3                0                   0                0            0
4                0                   0                1            0

   location_USA  education_level_Bachelor  education_level_Diploma  \
0             0                         1                        0
1             0                         1                        0
2             0                         0                        0
3             0                         0                        0
4             0                         1                        0

   education_level_High School  education_level_Master  education_level_PhD  \
0                            0                       0                    0
1                            0                       0                    0
2                            0                       0                    1
3                            0                       0                    1
4                            0                       0                    0

   remote_work_Hybrid  remote_work_No  remote_work_Yes
0                True           False            False
1               False            True            False
2               False            True            False
3               False           False             True
4               False           False             True
```

In [32]:
```
print(data.columns)
```

```
Index(['experience_years', 'skills_count', 'certifications', 'salary',
       'job_title_AI Engineer', 'job_title_Backend Developer',
       'job_title_Business Analyst', 'job_title_Cloud Engineer',
       'job_title_Cybersecurity Analyst', 'job_title_Data Analyst',
       'job_title_Data Scientist', 'job_title_DevOps Engineer',
       'job_title_Frontend Developer', 'job_title_Machine Learning Engineer',
       'job_title_Product Manager', 'job_title_Software Engineer',
       'industry_Consulting', 'industry_Education', 'industry_Finance',
       'industry_Government', 'industry_Healthcare', 'industry_Manufacturing',
       'industry_Media', 'industry_Retail', 'industry_Technology',
       'industry_Telecom', 'company_size_Enterprise', 'company_size_Large',
       'company_size_Medium', 'company_size_Small', 'company_size_Startup',
       'location_Australia', 'location_Canada', 'location_Germany',
       'location_India', 'location_Netherlands', 'location_Remote',
       'location_Singapore', 'location_Sweden', 'location_UK', 'location_USA',
       'education_level_Bachelor', 'education_level_Diploma',
       'education_level_High School', 'education_level_Master',
       'education_level_PhD', 'remote_work_Hybrid', 'remote_work_No',
       'remote_work_Yes'],
      dtype='str')
```

In [33]:

```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
data.columns = data.columns.str.replace(' ','_')
data = pd.get_dummies(data,columns=['job_title','industry','company_size','locat
pd.set_option('display.max_columns',None)
data[[col for col in data.columns if 'industry' in col]].head()
data[[col for col in data.columns if 'company_size'in col]].head()
print(data.head())
```

```
   experience_years  skills_count  certifications  salary  \
0                10             2               2  109413
1                 5            17               0   93764
2                18             4               1  148123
3                19            13               0  189123
4                15             7               0  165069

   job_title_AI Engineer  job_title_Backend Developer  \
0                      1                            0
1                      0                            0
2                      0                            0
3                      0                            0
4                      0                            0

   job_title_Business Analyst  job_title_Cloud Engineer  \
0                           0                         0
1                           0                         0
2                           0                         0
3                           1                         0
4                           0                         0

   job_title_Cybersecurity Analyst  job_title_Data Analyst  \
0                                0                       0
1                                0                       1
2                                0                       0
3                                0                       0
4                                0                       0

   job_title_Data Scientist  job_title_DevOps Engineer  \
0                         0                          0
1                         0                          0
2                         0                          0
3                         0                          0
4                         0                          0

   job_title_Frontend Developer  job_title_Machine Learning Engineer  \
0                             0                                    0
1                             0                                    0
2                             1                                    0
3                             0                                    0
4                             0                                    0

   job_title_Product Manager  job_title_Software Engineer  \
0                          0                            0
1                          0                            0
2                          0                            0
3                          0                            0
4                          1                            0

   industry_Consulting  industry_Education  industry_Finance  \
0                    0                   0                 0
1                    0                   0                 0
2                    0                   0                 0
3                    0                   0                 0
4                    0                   0                 0

   industry_Government  industry_Healthcare  industry_Manufacturing  \
0                    0                   1                       0
1                    0                   0                       0
2                    0                   0                       0
```

```
3                    0                    0                       0
4                    0                    0                       1

   industry_Media  industry_Retail  industry_Technology  industry_Telecom  \
0               0                0                    0                 0
1               0                0                    0                 1
2               1                0                    0                 0
3               0                1                    0                 0
4               0                0                    0                 0

   company_size_Enterprise  company_size_Large  company_size_Medium  \
0                        0                   0                    1
1                        0                   0                    0
2                        0                   0                    1
3                        0                   0                    1
4                        0                   1                    0

   company_size_Small  company_size_Startup  location_Australia  \
0                   0                     0                   0
1                   1                     0                   1
2                   0                     0                   0
3                   0                     0                   0
4                   0                     0                   0

   location_Canada  location_Germany  location_India  location_Netherlands  \
0                0                 0               1                     0
1                0                 0               0                     0
2                0                 0               0                     0
3                1                 0               0                     0
4                0                 0               0                     0

   location_Remote  location_Singapore  location_Sweden  location_UK  \
0                0                   0                0            0
1                0                   0                0            0
2                0                   1                0            0
3                0                   0                0            0
4                0                   0                1            0

   location_USA  education_level_Bachelor  education_level_Diploma  \
0             0                         1                        0
1             0                         1                        0
2             0                         0                        0
3             0                         0                        0
4             0                         1                        0

   education_level_High School  education_level_Master  education_level_PhD  \
0                            0                       0                    0
1                            0                       0                    0
2                            0                       0                    1
3                            0                       0                    1
4                            0                       0                    0

   remote_work_Hybrid  remote_work_No  remote_work_Yes
0                   1               0                0
1                   0               1                0
2                   0               1                0
3                   0               0                1
4                   0               0                1
```

In [34]:
```
print(data.shape)
```

```
(250000, 49)
```

In [35]:
```
print(data.columns)
```

```
Index(['experience_years', 'skills_count', 'certifications', 'salary',
       'job_title_AI Engineer', 'job_title_Backend Developer',
       'job_title_Business Analyst', 'job_title_Cloud Engineer',
       'job_title_Cybersecurity Analyst', 'job_title_Data Analyst',
       'job_title_Data Scientist', 'job_title_DevOps Engineer',
       'job_title_Frontend Developer', 'job_title_Machine Learning Engineer',
       'job_title_Product Manager', 'job_title_Software Engineer',
       'industry_Consulting', 'industry_Education', 'industry_Finance',
       'industry_Government', 'industry_Healthcare', 'industry_Manufacturing',
       'industry_Media', 'industry_Retail', 'industry_Technology',
       'industry_Telecom', 'company_size_Enterprise', 'company_size_Large',
       'company_size_Medium', 'company_size_Small', 'company_size_Startup',
       'location_Australia', 'location_Canada', 'location_Germany',
       'location_India', 'location_Netherlands', 'location_Remote',
       'location_Singapore', 'location_Sweden', 'location_UK', 'location_USA',
       'education_level_Bachelor', 'education_level_Diploma',
       'education_level_High School', 'education_level_Master',
       'education_level_PhD', 'remote_work_Hybrid', 'remote_work_No',
       'remote_work_Yes'],
      dtype='str')
```

In [36]:
```
print(data.isnull().sum())
```

```
experience_years                        0
skills_count                            0
certifications                          0
salary                                  0
job_title_AI Engineer                   0
job_title_Backend Developer             0
job_title_Business Analyst              0
job_title_Cloud Engineer                0
job_title_Cybersecurity Analyst         0
job_title_Data Analyst                  0
job_title_Data Scientist                0
job_title_DevOps Engineer               0
job_title_Frontend Developer            0
job_title_Machine Learning Engineer     0
job_title_Product Manager               0
job_title_Software Engineer             0
industry_Consulting                     0
industry_Education                      0
industry_Finance                        0
industry_Government                     0
industry_Healthcare                     0
industry_Manufacturing                  0
industry_Media                          0
industry_Retail                         0
industry_Technology                     0
industry_Telecom                        0
company_size_Enterprise                 0
company_size_Large                      0
company_size_Medium                     0
company_size_Small                      0
company_size_Startup                    0
location_Australia                      0
location_Canada                         0
location_Germany                        0
location_India                          0
location_Netherlands                    0
location_Remote                         0
location_Singapore                      0
location_Sweden                         0
location_UK                             0
location_USA                            0
education_level_Bachelor                0
education_level_Diploma                 0
education_level_High School             0
education_level_Master                  0
education_level_PhD                     0
remote_work_Hybrid                      0
remote_work_No                          0
remote_work_Yes                         0
dtype: int64
```

In [37]:

```
import pandas as pd
data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data
data['education_level'] = data['education_level'].astype('category').cat.codes
data = pd.get_dummies(data,columns=['job_title','industry','location','remote_wo
print(data.head())
```

```
   experience_years  education_level  skills_count  certifications  salary  \
0                10                0             2               2  109413
1                 5                0            17               0   93764
2                18                4             4               1  148123
3                19                4            13               0  189123
4                15                0             7               0  165069

   job_title_AI Engineer  job_title_Backend Developer  \
0                      1                            0
1                      0                            0
2                      0                            0
3                      0                            0
4                      0                            0

   job_title_Business Analyst  job_title_Cloud Engineer  \
0                           0                         0
1                           0                         0
2                           0                         0
3                           1                         0
4                           0                         0

   job_title_Cybersecurity Analyst  job_title_Data Analyst  \
0                                0                       0
1                                0                       1
2                                0                       0
3                                0                       0
4                                0                       0

   job_title_Data Scientist  job_title_DevOps Engineer  \
0                         0                          0
1                         0                          0
2                         0                          0
3                         0                          0
4                         0                          0

   job_title_Frontend Developer  job_title_Machine Learning Engineer  \
0                             0                                    0
1                             0                                    0
2                             1                                    0
3                             0                                    0
4                             0                                    0

   job_title_Product Manager  job_title_Software Engineer  \
0                          0                            0
1                          0                            0
2                          0                            0
3                          0                            0
4                          1                            0

   industry_Consulting  industry_Education  industry_Finance  \
0                    0                   0                 0
1                    0                   0                 0
2                    0                   0                 0
3                    0                   0                 0
4                    0                   0                 0

   industry_Government  industry_Healthcare  industry_Manufacturing  \
0                    0                   1                       0
1                    0                   0                       0
2                    0                   0                       0
```

```
3                    0                    0                       0
4                    0                    0                       1

   industry_Media  industry_Retail  industry_Technology  industry_Telecom  \
0               0                0                    0                 0
1               0                0                    0                 1
2               1                0                    0                 0
3               0                1                    0                 0
4               0                0                    0                 0

   location_Australia  location_Canada  location_Germany  location_India  \
0                   0                0                 0               1
1                   1                0                 0               0
2                   0                0                 0               0
3                   0                1                 0               0
4                   0                0                 0               0

   location_Netherlands  location_Remote  location_Singapore  location_Sweden  \
0                     0                0                   0                0
1                     0                0                   0                0
2                     0                0                   1                0
3                     0                0                   0                0
4                     0                0                   0                1

   location_UK  location_USA  remote_work_Hybrid  remote_work_No  \
0            0             0                   1               0
1            0             0                   0               1
2            0             0                   0               1
3            0             0                   0               0
4            0             0                   0               0

   remote_work_Yes  company_size_Enterprise  company_size_Large  \
0                0                        0                   0
1                0                        0                   0
2                0                        0                   0
3                1                        0                   0
4                1                        0                   1

   company_size_Medium  company_size_Small  company_size_Startup
0                    1                   0                     0
1                    0                   1                     0
2                    1                   0                     0
3                    1                   0                     0
4                    0                   0                     0
```

In [38]:
```
print(data.shape)
```

```
(250000, 45)
```

In [39]:
```
print(data.columns)
```

```
Index(['experience_years', 'education_level', 'skills_count', 'certifications',
       'salary', 'job_title_AI Engineer', 'job_title_Backend Developer',
       'job_title_Business Analyst', 'job_title_Cloud Engineer',
       'job_title_Cybersecurity Analyst', 'job_title_Data Analyst',
       'job_title_Data Scientist', 'job_title_DevOps Engineer',
       'job_title_Frontend Developer', 'job_title_Machine Learning Engineer',
       'job_title_Product Manager', 'job_title_Software Engineer',
       'industry_Consulting', 'industry_Education', 'industry_Finance',
       'industry_Government', 'industry_Healthcare', 'industry_Manufacturing',
       'industry_Media', 'industry_Retail', 'industry_Technology',
       'industry_Telecom', 'location_Australia', 'location_Canada',
       'location_Germany', 'location_India', 'location_Netherlands',
       'location_Remote', 'location_Singapore', 'location_Sweden',
       'location_UK', 'location_USA', 'remote_work_Hybrid', 'remote_work_No',
       'remote_work_Yes', 'company_size_Enterprise', 'company_size_Large',
       'company_size_Medium', 'company_size_Small', 'company_size_Startup'],
      dtype='str')
```

In [40]:
```
print(data.head())
```

```
   experience_years  education_level  skills_count  certifications  salary  \
0                10                0             2               2  109413
1                 5                0            17               0   93764
2                18                4             4               1  148123
3                19                4            13               0  189123
4                15                0             7               0  165069

   job_title_AI Engineer  job_title_Backend Developer  \
0                      1                            0
1                      0                            0
2                      0                            0
3                      0                            0
4                      0                            0

   job_title_Business Analyst  job_title_Cloud Engineer  \
0                           0                         0
1                           0                         0
2                           0                         0
3                           1                         0
4                           0                         0

   job_title_Cybersecurity Analyst  job_title_Data Analyst  \
0                                0                       0
1                                0                       1
2                                0                       0
3                                0                       0
4                                0                       0

   job_title_Data Scientist  job_title_DevOps Engineer  \
0                         0                          0
1                         0                          0
2                         0                          0
3                         0                          0
4                         0                          0

   job_title_Frontend Developer  job_title_Machine Learning Engineer  \
0                             0                                    0
1                             0                                    0
2                             1                                    0
3                             0                                    0
4                             0                                    0

   job_title_Product Manager  job_title_Software Engineer  \
0                          0                            0
1                          0                            0
2                          0                            0
3                          0                            0
4                          1                            0

   industry_Consulting  industry_Education  industry_Finance  \
0                    0                   0                 0
1                    0                   0                 0
2                    0                   0                 0
3                    0                   0                 0
4                    0                   0                 0

   industry_Government  industry_Healthcare  industry_Manufacturing  \
0                    0                   1                       0
1                    0                   0                       0
2                    0                   0                       0
```

```
3                    0                    0                       0
4                    0                    0                       1

   industry_Media  industry_Retail  industry_Technology  industry_Telecom  \
0               0                0                    0                 0
1               0                0                    0                 1
2               1                0                    0                 0
3               0                1                    0                 0
4               0                0                    0                 0

   location_Australia  location_Canada  location_Germany  location_India  \
0                   0                0                 0               1
1                   1                0                 0               0
2                   0                0                 0               0
3                   0                1                 0               0
4                   0                0                 0               0

   location_Netherlands  location_Remote  location_Singapore  location_Sweden  \
0                     0                0                   0                0
1                     0                0                   0                0
2                     0                0                   1                0
3                     0                0                   0                0
4                     0                0                   0                1

   location_UK  location_USA  remote_work_Hybrid  remote_work_No  \
0            0             0                   1               0
1            0             0                   0               1
2            0             0                   0               1
3            0             0                   0               0
4            0             0                   0               0

   remote_work_Yes  company_size_Enterprise  company_size_Large  \
0                0                        0                   0
1                0                        0                   0
2                0                        0                   0
3                1                        0                   0
4                1                        0                   1

   company_size_Medium  company_size_Small  company_size_Startup
0                    1                   0                     0
1                    0                   1                     0
2                    1                   0                     0
3                    1                   0                     0
4                    0                   0                     0
```

In [41]:
```
print(data.isnull().sum())
```

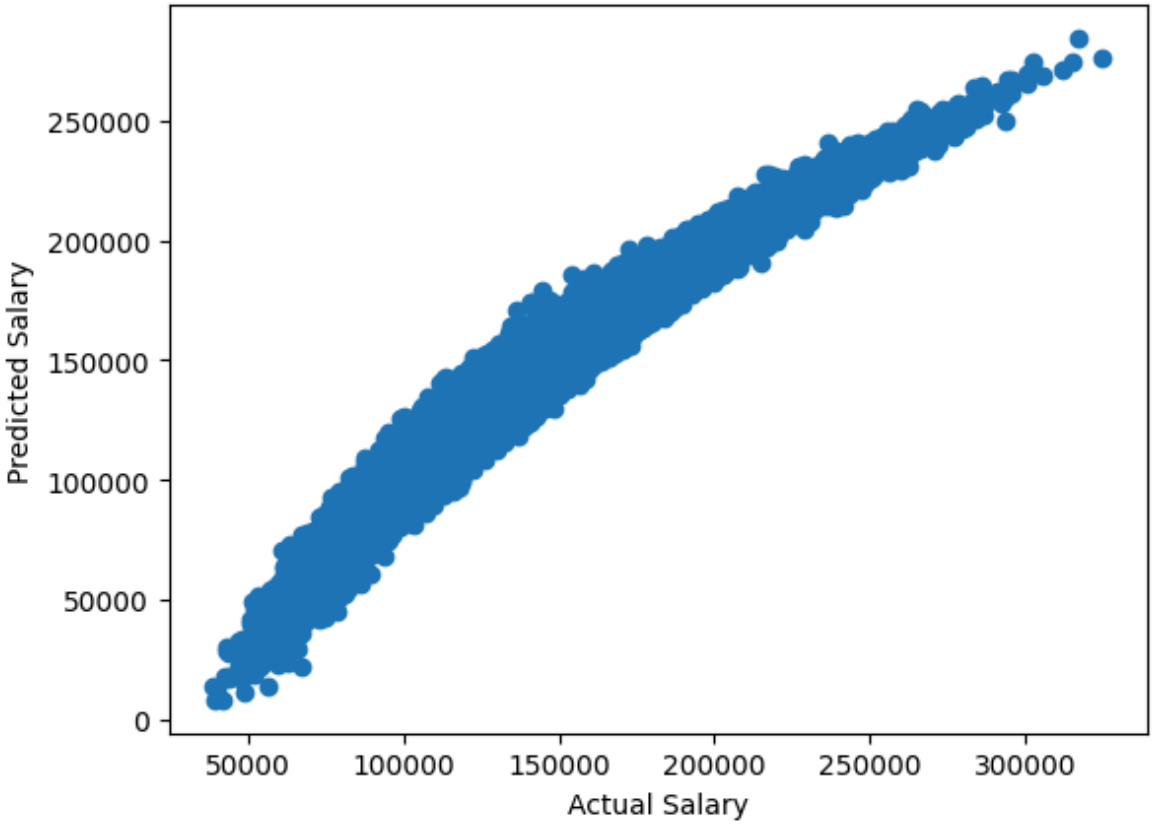
```
experience_years                        0
education_level                         0
skills_count                            0
certifications                          0
salary                                  0
job_title_AI Engineer                   0
job_title_Backend Developer             0
job_title_Business Analyst              0
job_title_Cloud Engineer                0
job_title_Cybersecurity Analyst         0
job_title_Data Analyst                  0
job_title_Data Scientist                0
job_title_DevOps Engineer               0
job_title_Frontend Developer            0
job_title_Machine Learning Engineer     0
job_title_Product Manager               0
job_title_Software Engineer             0
industry_Consulting                     0
industry_Education                      0
industry_Finance                        0
industry_Government                     0
industry_Healthcare                     0
industry_Manufacturing                  0
industry_Media                          0
industry_Retail                         0
industry_Technology                     0
industry_Telecom                        0
location_Australia                      0
location_Canada                         0
location_Germany                        0
location_India                          0
location_Netherlands                    0
location_Remote                         0
location_Singapore                      0
location_Sweden                         0
location_UK                             0
location_USA                            0
remote_work_Hybrid                      0
remote_work_No                          0
remote_work_Yes                         0
company_size_Enterprise                 0
company_size_Large                      0
company_size_Medium                     0
company_size_Small                      0
company_size_Startup                    0
dtype: int64
```

In [42]:

```
import pandas as pd

data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data

data = pd.get_dummies(data,dtype=int)

X = data.drop('salary', axis=1)
y = data['salary']

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_

from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

from sklearn.metrics import mean_absolute_error, mean_squared_error
print("MAE:", mean_absolute_error(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))

import matplotlib.pyplot as plt
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.show()
```

```
MAE: 5436.096936751799
MSE: 50773076.6572084
```

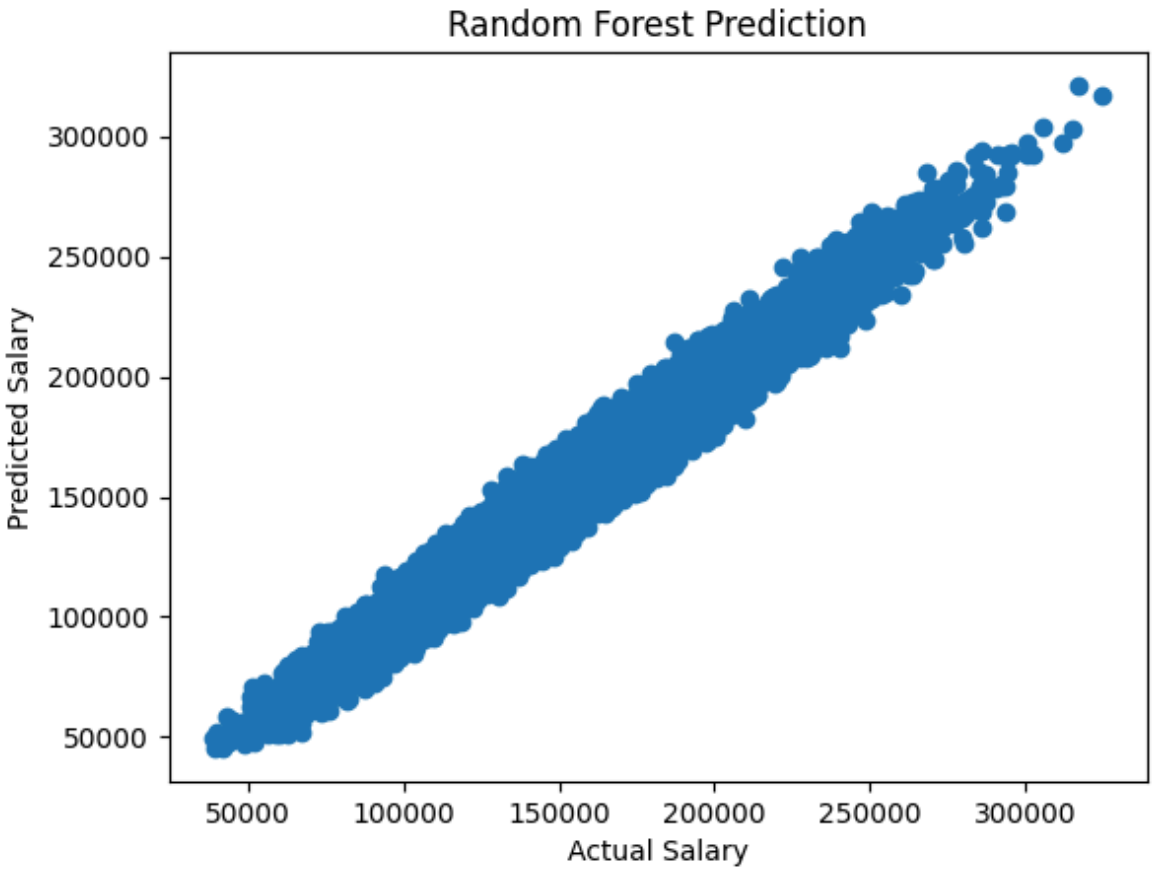


In [48]:

```
import pandas as pd

data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data

data = pd.get_dummies(data,dtype=int)
X = data.drop('salary', axis=1)
y = data['salary']

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_

from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor()
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

print("MAE:", mean_absolute_error(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

import matplotlib.pyplot as plt

plt.scatter(y_test, y_pred)
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.title("Random Forest Prediction")
plt.show()
```

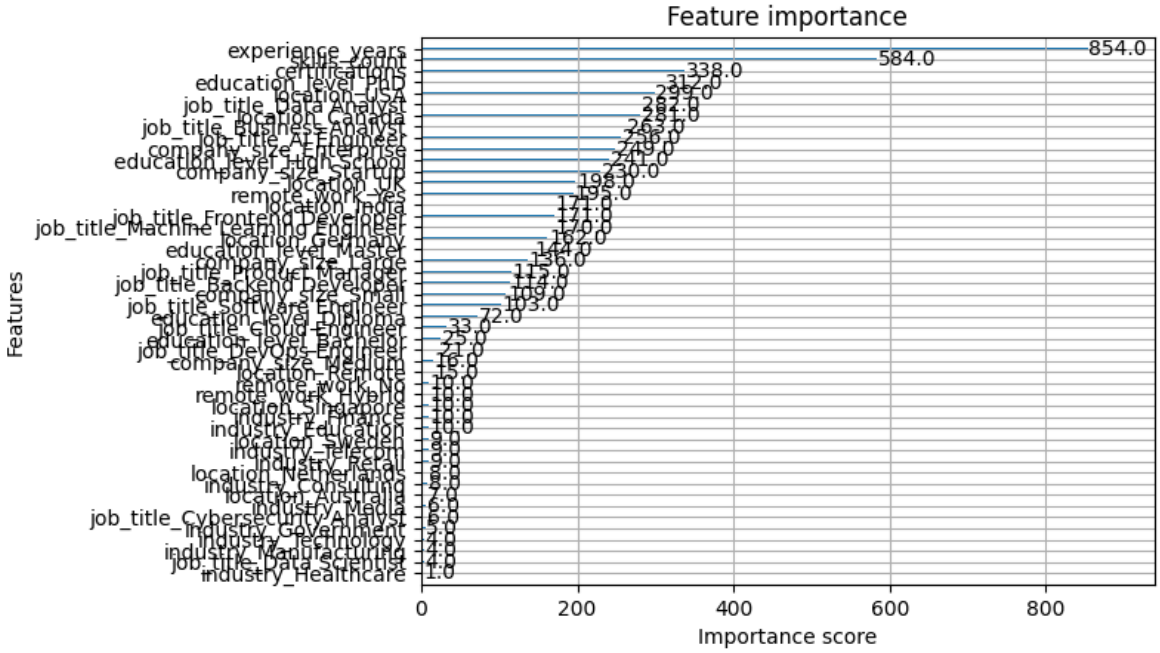
```
MAE: 5026.307788019523
MSE: 40085608.247890316
R2 Score: 0.9711586034258525
```

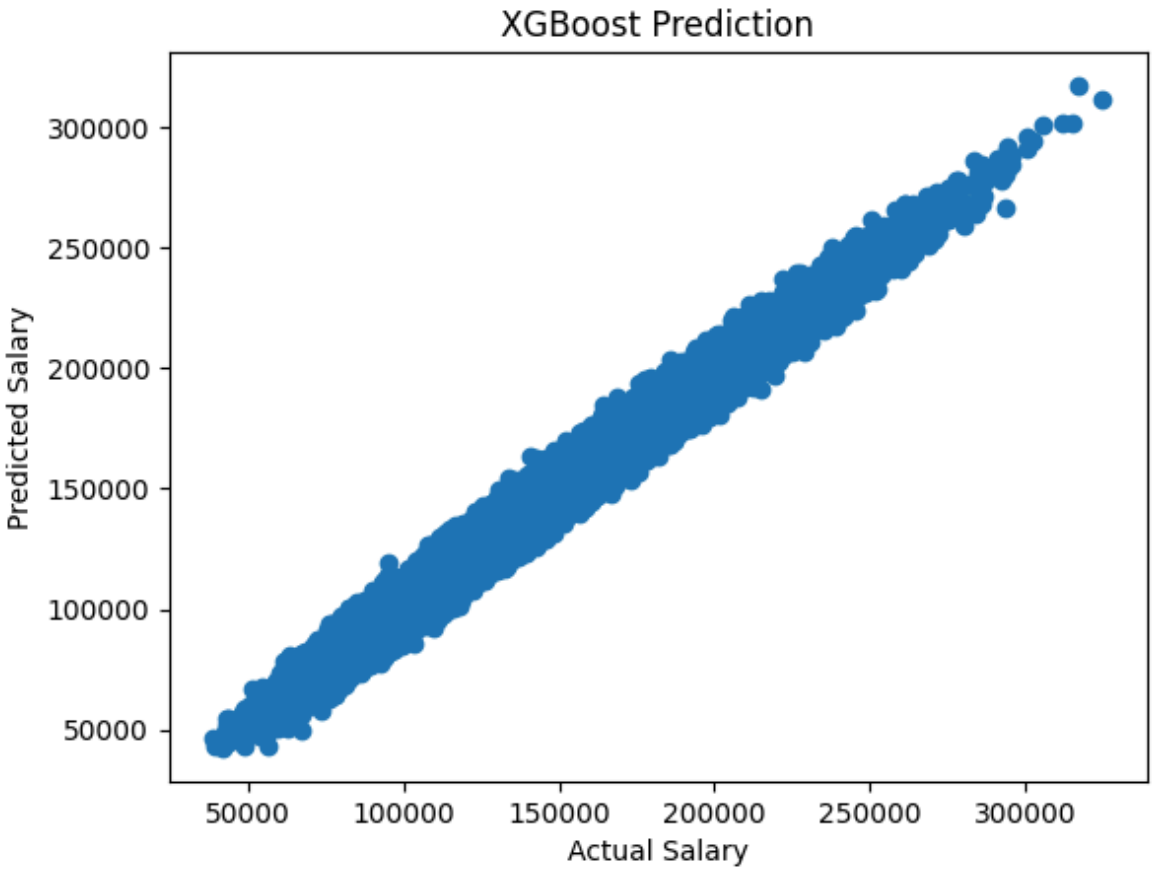


In [2]:
```
import pandas as pd

data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data

data = pd.get_dummies(data,dtype=int)
X = data.drop('salary', axis=1)
y = data['salary']

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_

from xgboost import XGBRegressor
model = XGBRegressor(n_estimators=100, learning_rate=0.1)
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)

from sklearn.metrics import r2_score

print("R2 Score:", r2_score(y_test, y_pred))
import matplotlib.pyplot as plt
from xgboost import plot_importance

plot_importance(model)
plt.show()
import matplotlib.pyplot as plt

plt.scatter(y_test, y_pred)
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.title("XGBoost Prediction")
plt.show()
```

```
R2 Score: 0.9785695672035217
```





In [4]:
```
import pandas as pd

data = pd.read_csv(r"C:\Users\satya\Downloads\archive\job_salary_prediction_data

data = pd.get_dummies(data,dtype=int)
X = data.drop('salary', axis=1)
y = data['salary']

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_

from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import r2_score

# Linear Regression
lr = LinearRegression()
lr.fit(X_train, y_train)
lr_pred = lr.predict(X_test)

# Random Forest
rf = RandomForestRegressor()
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)

# XGBoost
xgb = XGBRegressor(n_estimators=100, learning_rate=0.1)
xgb.fit(X_train, y_train)
xgb_pred = xgb.predict(X_test)

# Print results
```

```
print("Linear R2:", r2_score(y_test, lr_pred))
print("Random Forest R2:", r2_score(y_test, rf_pred))
print("XGBoost R2:", r2_score(y_test, xgb_pred))
```

```
Linear R2: 0.9634690228446962
Random Forest R2: 0.9712192991550164
XGBoost R2: 0.9785695672035217
```

In [ ]: